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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: PROJECT EVALUATION (24CS2235F) A.Y. 2026 - 2027

TITLE OF THE PROJECT	**Lexical Analyzer Using LEX Tool**	
S. No.	**University ID**	**Name**
1	2420030257	HANEESH
2	2420030465	ASHISH
3	2420030455	HRISHIKESH
4	2420090049	ARYAN
Name of the Guide		Mr B. Dwarakanath

Abstract

The Smart Lexical Analyzer is a compiler-design project developed to analyze source code and convert it into a structured sequence of tokens. The system identifies keywords, identifiers, operators, literals, delimiters, and other lexical elements while tracking their line and column positions. Unlike a basic lexical analyzer that only displays token output, the proposed system provides token visualization, lexical error detection, and automatic symbol table generation. It identifies common lexical errors such as invalid identifiers, malformed numeric literals, unterminated strings, and unrecognized characters, while allowing the analysis to continue after an error is encountered. The generated symbol table stores relevant information such as identifier names, data types, values, line numbers, and scope. A visual interface presents the analysis results in an organized manner, making the system useful for understanding the lexical analysis phase of a compiler and demonstrating practical applications of finite automata and regular expressions. The system also provides a concise analysis summary, helping users understand the structure and characteristics of the analyzed source code. Overall, the project combines theoretical compiler concepts with an interactive approach to make lexical analysis easier to understand and interpret.

Signature of the Guide HOD